

Auditing HeSum: Reference Quality Defects in a Hebrew News Summarization Corpus

<https://github.com/avrymi-asraf/AMLK-the-true-title>

Amit Benbenishti

amit.benbenishti@mail.huji.ac.il

Avraham Asraf

avraham.asraf@mail.huji.ac.il

Ofek Varona

ofek.varona@mail.huji.ac.il

Abstract

HeSum is one of the few corpora large enough to train modern Hebrew news summarizers, but its targets are extended subheadings scraped from news sites rather than summaries written for modeling. We audit whether those targets are suitable references before treating them as ground truth. A reproducible pipeline filters unusable rows and rewrites defective headlines; an independent rubric judge then scores original and curated headlines directly against their articles for faithfulness, single-focus, informativeness, and cleanliness. The audit removes 41.5% of the source corpus and rewrites 52.4% of the retained headlines. Flagged references show large, defect-specific quality losses, while curated headlines are preferred in 73.6% of blind comparisons. These results show that scraped reference quality is a first-order experimental variable, not a fixed property of a benchmark. A matched fine-tuning comparison likewise improves model-output faithfulness and informativeness. We release the resulting audit and frame target validation as a prerequisite for trustworthy Hebrew summarization training and evaluation.

1 Introduction

HeSum (Paz-Argaman et al., 2024) is one of the few public resources large enough to fine-tune a modern Hebrew news summarizer. Its “summaries,” however, are extended subheadings scraped from news sites, not references written for the task. Such targets can bundle several stories, contain boilerplate, or assert details absent from the paired article. Training and evaluating against them as if they were gold therefore risks rewarding a model for reproducing dataset defects.

Current summarization practice usually fine-tunes on the available corpus and reports overlap metrics such as ROUGE or BERTScore (Zhang et al., 2020), both of which assume the reference is trustworthy – an assumption especially fragile

for scraped targets, where lead-derived references can reward copying the opening sentence rather than summarizing (Nallapati et al., 2016; See et al., 2017).

The problem became visible during qualitative inspection of model outputs. Fluent generated headlines were sometimes more focused and better grounded than the HeSum headline used to score them. Returning to the raw corpus revealed recurring, source-driven patterns rather than isolated annotation mistakes: pipe-delimited digests joined unrelated headlines, roundup pages paired one article body with claims about several stories, and extraction artifacts survived in both article text and target. These cases made a model-only diagnosis inadequate. If the target itself is defective, improving architecture or optimization cannot make reference-based evaluation trustworthy.

We reverse the usual order: audit the references first, then train. We contribute (i) a reproducible pipeline that filters unusable HeSum rows and repairs defective headlines, (ii) an article-grounded, four-dimension rubric that evaluates references without requiring a second reference string, and (iii) independent graded and pairwise tests of whether the flagged rows are genuinely worse and the rewrites genuinely better. The audit is the main claim; E4 applies the same rubric and blind comparison to models fine-tuned on uncleaned versus curated data.

2 Data

HeSum (Paz-Argaman et al., 2024) contains 10,000 real-world Hebrew news article–headline pairs. Our pipeline first trims repeated byline and related-link tails from 722 articles. Two deterministic filters then run independently: the token-budget filter flags 2,659 rows whose article and

headline exceed 4,000 DictaLM tokens, and the multi-pipe filter flags 2,412 headlines containing at least two pipe separators. Their intersection leaves 6,486 rows before any model decision is introduced.

An LLM curator (gpt-5.6-luna) assigns each survivor one source-usability label with explicit negative guards: usable, multiple independent items, insufficient substantive content, substantive content absent from the text, damaged or fragmentary text, or other. It removes 632 unusable rows, then either keeps or rewrites each usable headline. This separation matters: deterministic exclusions do not depend on the curator, and source usability is decided before headline wording is repaired. The final corpus contains 5,854 rows, of which 2,785 retain their original headline and 3,069 receive a rewrite. Figure 1 gives the complete flow and Table 1 defines the overlapping analysis groups used below.

The stratum numbering deliberately omits S1 (over-token-budget). Because the filter is a hard length cut, S1 and clean rows have no common support in article length and cannot support a valid group comparison. We instead analyze length continuously as a supplementary lead-bias check (Appendix Figure 8). The curated corpus splits 80/10/10 by article into train/validation/test (4,683/585/586 rows); Section 3 describes how E4’s two training arms and shared test subset use this split.

3 Methods

Article-grounded rubric. String-overlap metrics cannot establish whether a lone reference is correct. We therefore use Gemini gemini-2.5-flash-lite to read the article and score a headline from 1–5 on *faithfulness* (every claim is supported), *single-focus* (one dominant event rather than a bundle), *informativeness* (the reader can identify what happened), and *cleanliness* (no scraping or formatting artifacts). Keeping the dimensions separate prevents a structurally messy headline from being treated as factually wrong, and lets each filter predict a specific failure signature.

The judge is blind to stratum and provenance: it never sees whether a headline is original, curated, or expected to fail. It also belongs to a different model family from the curator, reducing self-preference risk (Panickssery et al., 2024).

Twelve Hebrew worked examples—scores 1, 3, and 5 for each dimension—anchor the ordinal scale and are excluded from downstream analysis. On the pilot, quadratically weighted test–retest agreement is 0.86 for faithfulness, 0.90 for single-focus, 0.76 for informativeness, and 0.65 for cleanliness; a three-annotator human validation round is reported in Section 4 and Figure 5.

Four pre-registered experiments use this instrument to answer four separate questions: are the flagged strata actually worse (E1); does repairing a headline improve its own score (E2); do blind readers agree (E3); and does training on the curated corpus rather than the uncleaned data change a model’s output (E4).

E1: Are the flagged strata worse than clean headlines? E1 scores 9,988 original headlines and asks whether each flagged stratum scores lower than clean S0 on the rubric, with a pre-registered prediction for which dimension each defect should hurt most (e.g. multi-pipe rows should primarily lower single-focus). Results use Cliff’s δ and bootstrap confidence intervals (Cliff, 1993).

E2: Does the paired repair raise a headline’s own score? E2 compares original and curated scores on 3,068 rewritten pairs and reports the proportions that improve, tie, or regress.

E3: Do blind readers prefer the repaired headline? Because a five-point scale compresses changes near its ceiling, E3 presents both headlines in randomized order to a blind pairwise judge and checks position bias.

E4: Does training on the curated corpus change model quality? Two arms fine-tune dicta-il/dictalm2.0-instruct with LoRA (Hu et al., 2022) for one epoch under matched optimization on equal-size (4,683-row) training sets: Arm A on uncleaned HeSum and Arm B on the curated corpus. Curated validation/test IDs are excluded from Arm A to prevent leakage. Both arms generate on the same seeded 120-article test subset, scored with the E1–E3 rubric and a blind pairwise judge (Section 4).

4 Results

Flagged references are genuinely worse (E1). Flagged strata score lower than clean S0 in every pre-registered direction (Figure 2). The clearest case is S2 multi-pipe on single-focus ($\delta = -0.95$,

median 1 vs. 5): pipe-delimited headlines bundle several stories instead of one. S2 and S3 also lose faithfulness and informativeness; S4 originals stay near S0 ($|\delta| \leq 0.27$), where the curator rewrote stylistically weak lines the judge still scores as acceptable (Figure 3).

The rewrites improve the target (E2/E3). On 3,068 paired rewrites, faithfulness increases for 42.0% of rows and decreases for 3.6%; the transition matrix shows that movement is overwhelmingly upward (Figure 4). Because medians remain compressed at the top of the five-point scale, the blind comparison provides the clearer magnitude: curated headlines win 73.6% of rewritten comparisons ($n = 961$, 95% CI [70.8%, 76.4%]; Figure 6). The benefit is concentrated in substantive edits: full rewrites win 76.8% of comparisons ($n = 833$), whereas light edits are near a coin flip (49.5%, $n = 97$). Thus curation is not uniformly beneficial merely because text changed; it helps most where the original target required reconstruction.

E1, E2, and E3 answer different questions – validating the labels, tracking which dimensions move after repair, and measuring preference without ceiling compression – and their agreement supports a narrow claim: the pipeline identifies real reference defects and improves the headlines it rewrites. E4 asks whether that target-side improvement shows up in model outputs under matched fine-tuning.

Humans corroborate the automated judge. Three annotators independently scored disjoint subsets of a stratified worklist, pooling to $n = 152$ rubric articles and $n = 95$ blind pairwise comparisons (no overlapping rows, so inter-annotator reliability is not estimable). Human and judge scores agree within one ordinal point on at least 68% of articles per dimension (Figure 5) – cleanliness 90%, single-focus 86%, informativeness 76%, faithfulness 68%. On the $n = 78$ non-tie pairwise comparisons, using the same non-tie win-rate convention as E3, humans prefer the curated headline 83.3% of the time, at least as decisively as the automated judge’s 73.6% win rate.

Training on the curated corpus improves model outputs (E4). On the shared $n = 120$ seeded test subset, curated-SFT raises mean faithfulness from 3.60 to 4.09 and informativeness from 3.90 to 4.33. Faithfulness improves on 39% of articles

(worsens on 18%); informativeness improves on 38% (17% worsen) – the same dimensions E2 showed benefit from headline repair. Uncleaned-SFT leaves 36% of outputs at faithfulness ≤ 2 versus 21% after curated-SFT, with the score-5 share rising from 38% to 54% (Figure 7). Single-focus is already near ceiling on both arms (3.85 to 3.96); cleanliness trends lower (3.12 to 2.93) without a reliable shift, as more curated outputs land at mid-range clutter scores even while faithfulness rises. Blind pairwise preference favors curated outputs on 64.3% of non-tie comparisons (74/41/5 ties; Wilson 95% CI [55.3%, 72.5%]), consistent with rubric gains on faithfulness and informativeness rather than a uniform lift across all four dimensions. Together, E4 shows that cleaning training targets transfers to model behavior: outputs become more faithful and informative without the pairwise judge needing to know which arm was trained on curated data.

5 Limitations and Conclusion

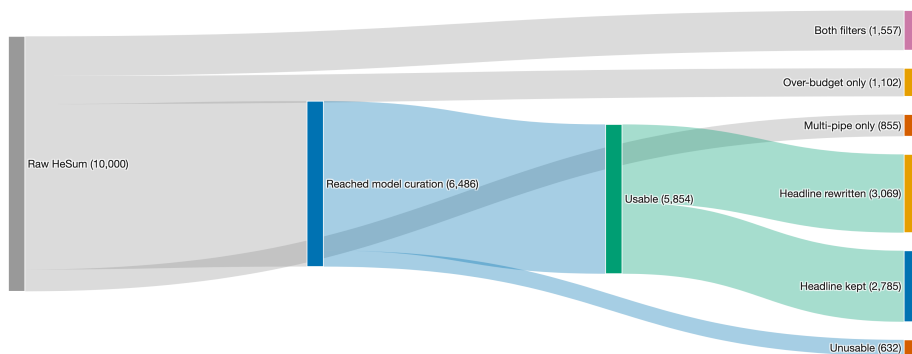
This audit shows that HeSum’s scraped headlines cannot be treated as fixed ground truth: reference quality varies systematically with observable extraction patterns (E1) and repair measurably improves it (E2/E3). This can reverse how a model score should be read – matching a bundled or unsupported headline more closely is not necessarily better summarization, so evaluation should ask whether the target is faithful before asking whether an output resembles it. The human validation round (Figure 5) corroborates the judge on non-overlapping slices but does not estimate inter-annotator reliability.

Two caveats bound this claim: curator and judge share no human gold standard, and the curator’s editorial style shapes what counts as a repair – it rewrote many headlines already scored near clean (S4), so a preferred rewrite is not proof of a unique target. E4’s fixed $n = 120$ subset (Figure 7) favors curated training, but its arms use different training pools and therefore do not isolate target repair from corpus composition. The audit’s claim does not depend on E4 – a scraped target is an observation, not a given.

A Figures and Tables

What the curation pipeline did

10,000 raw HeSum rows, through both deterministic filters, to the final usable/rewritten split



Source: data_curation pipeline - n = 10,000 raw HeSum rows

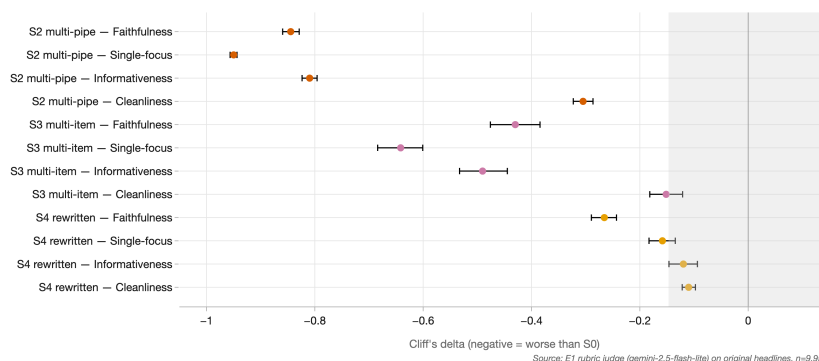
Figure 1: Curation funnel. Deterministic filters run independently before model labeling; 5,854 rows remain and 3,069 retained headlines are rewritten.

Stratum	<i>n</i>	Definition
S0 clean	2,785	passed every filter; headline kept
S2 multi-pipe	2,412	at least two pipe separators
S3 multi-item	569	multiple independent stories
S4 rewritten	3,069	curator replaced headline
S5 other unusable	63	collapsed low-frequency labels

Table 1: Overlapping analysis strata. S0 is the E1 reference group. S1 is omitted because the hard token threshold has no common length support with S0.

Effect sizes: Cliff's delta vs. S0 clean

95% bootstrap CI; shaded band is the negligible-effect range ($|\delta| < 0.147$)



Source: E1 rubric judge (gemini-2.5-flash-lite) on original headlines, n=9,958

Figure 2: E1 effect sizes relative to S0 clean. Negative Cliff's δ indicates lower reference quality; intervals are 95% bootstrap confidence intervals.

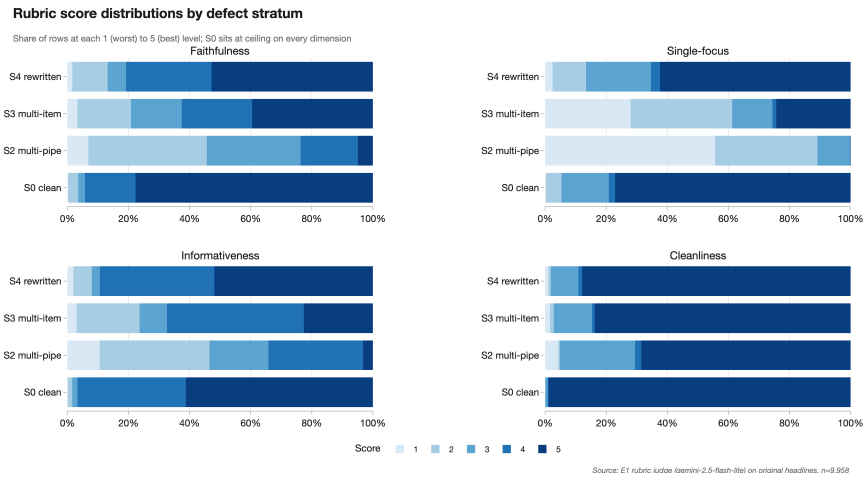


Figure 3: E1 rubric score distributions by stratum. S0 sits at ceiling on every dimension; S2 and S3 spread across the full range rather than shifting uniformly downward.

Original vs. curated faithfulness score, rewritten rows only

Off-diagonal mass below the diagonal would mean curation made a row worse -- it is rare

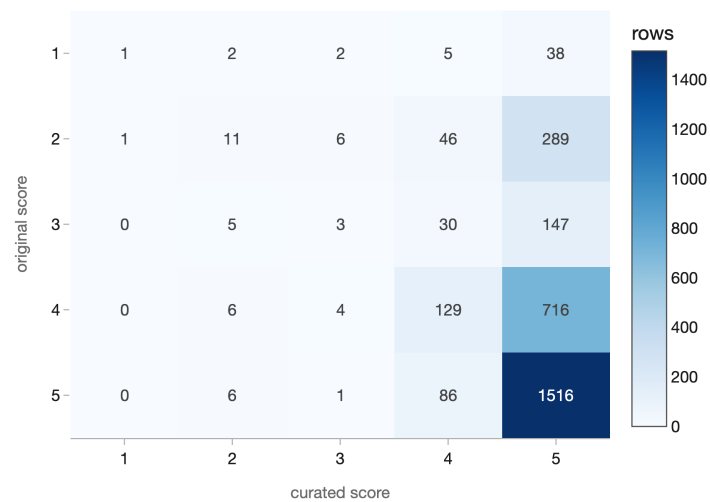


Figure 4: E2 original-to-curated faithfulness transitions. Most off-diagonal mass moves toward higher scores; regressions are rare.

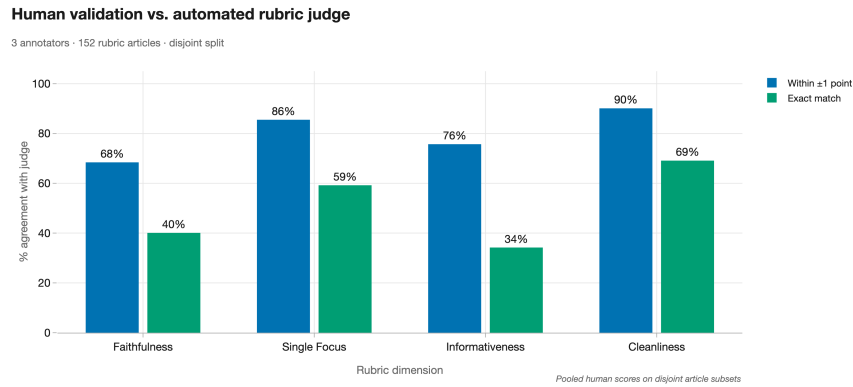


Figure 5: Human validation ($n = 152$ rubric articles, three annotators on disjoint subsets). Share of human–judge rubric pairs within one ordinal point and with an exact score match per dimension. Blind pairwise preference is reported in the text and in Figure 6.

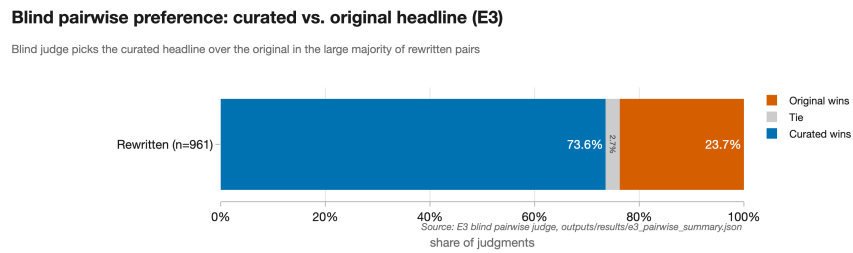


Figure 6: E3 blind preference. Curated headlines win 73.6% of rewritten comparisons against the original (95% CI [70.8%, 76.4%]).

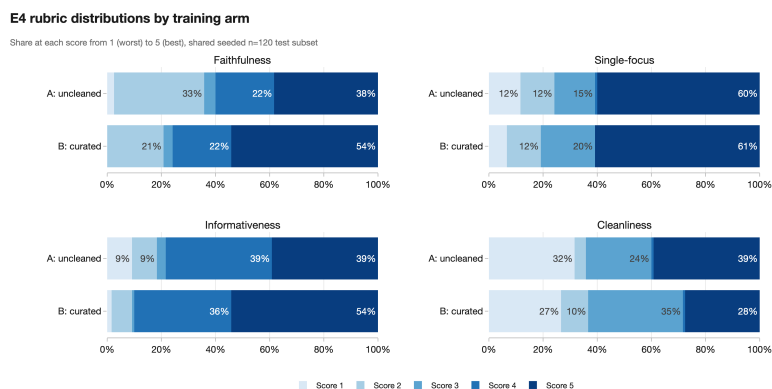


Figure 7: E4 rubric distributions on the shared seeded $n = 120$ test subset. Curated training shifts faithfulness and informativeness upward; single-focus is similar, while cleanliness trends lower without a reliable difference.

B Supplementary Lead-Bias Check

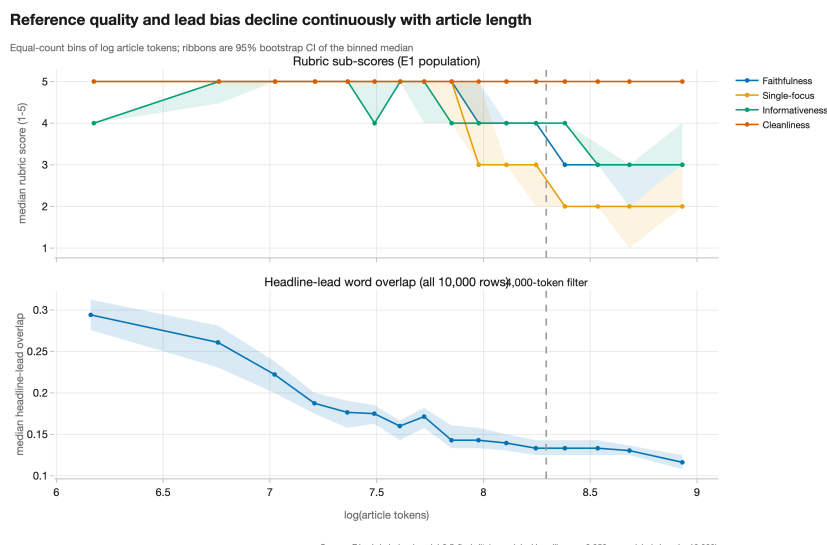


Figure 8: Supplementary analysis of rubric scores and headline–lead overlap across article length. The continuous trend motivates treating length as a covariate rather than comparing S1 directly with clean rows.

References

- Norman Cliff. 1993. Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin*, 114(3):494–509.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Ramesh Nallapati, Bowen Zhou, Cicero dos Santos, Caglar Gulcehre, and Bing Xiang. 2016. Abstractive text summarization using sequence-to-sequence rnns and beyond. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning*.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*.
- Tzuf Paz-Argaman, Itai Mondshine, Asaf Achi Mordechai, and Reut Tsarfaty. 2024. HeSum: a novel dataset for abstractive text summarization in hebrew. In *Findings of the Association for Computational Linguistics: ACL 2024*.
- Abigail See, Peter J. Liu, and Christopher D. Manning. 2017. Get to the point: Summarization with pointer-generator networks. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*.